

Vectors and Scalars

Definitions

Scalar quantities can be described by a single piece of information – their size

Examples include: *speed, distance*

Vector quantities need to have two pieces of information, their size and their direction

Examples include: **velocity, displacement**

Notation

Quite confused!

Arrow notation:

$\vec{a}$

for the Vector defined as a

or

$\overrightarrow{AB}$

for the Vector that takes
you from point A to point B

Bolded:

a

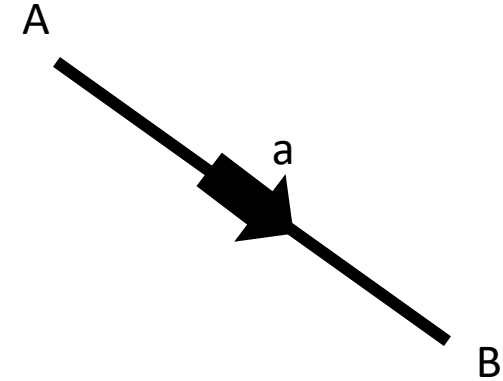
Much easier than arrows
to type, but can cause
confusion (esp. by hand!)

Underscored:

a

Clearer than bolding in
hand written work

$$\mathbf{a} = \overrightarrow{AB} = \underline{a}$$



Unit vectors

Notation: $\hat{i}, \hat{j}, \hat{k},$

Magnitude is defined as 1

Assigned to cardinal (orthogonal) dimensions (e.g. x, y, z)

Lets us define a vector as $\mathbf{a} = 2\hat{i} + 3\hat{j} - \hat{k}$

for the coordinates $x=2, y=3, z=-1$

These are Cartesian coordinates – we'll look at Polar form in a later class

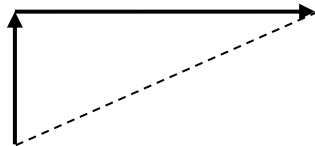
Addition

Adding two scalars is easy, they only have magnitude, so they add like regular numbers

$$1+2=3$$

Vectors have two pieces of information to include, magnitude and direction:

1 up + 2 right=



This fundamentally changes how we treat them

Subtraction

Subtracting a number is the same as adding a negative number

$$10-6=4$$

$$10+(-6)=4$$

If someone runs 5 miles from home and then 5 miles back home, their scalar distance is 10 miles (scalars don't have a direction, only a size!)

Their **vector displacement** is zero, since the +5 is cancelled out by the -5 (the same is true whatever the direction measure, if they run five miles North and then 5 miles South etc.)

Example of vector addition and subtraction

Given $\mathbf{a} = 7i + 3j$, $\mathbf{b} = i - 2j$, $\mathbf{c} = -3i + j$

What is $\mathbf{a} + \mathbf{b}$, what is $\mathbf{a} + \mathbf{b} + \mathbf{c}$ and what is $\mathbf{a} - \mathbf{c}$?

$$\mathbf{a} + \mathbf{b} = 7i + 3j + i - 2j = 8i + j$$

$$\mathbf{a} + \mathbf{b} + \mathbf{c} = 7i + 3j + i - 2j - 3i + j = 5i + 2j$$

$$\mathbf{a} - \mathbf{c} = 7i + 3j + 3i - j = 10i + 2j$$

Multiplication of vector and scalar

Multiplying a vector by a scalar changes the magnitude but not the direction

e.g. $\mathbf{a} = (2,3)$ so $\mathbf{a}$ is a vector that progresses 2 along the x axis and 3 along the y. This means it must have a magnitude of $\sqrt{2^2 + 3^2} \approx 3.6$

$2\mathbf{a} = (4,6)$, so $2\mathbf{a}$ is a vector that progresses 4 along the x axis and 6 along the y. This means it must have a magnitude of $\sqrt{4^2 + 6^2} \approx 7.2$

Multiplication of two vectors – the dot product

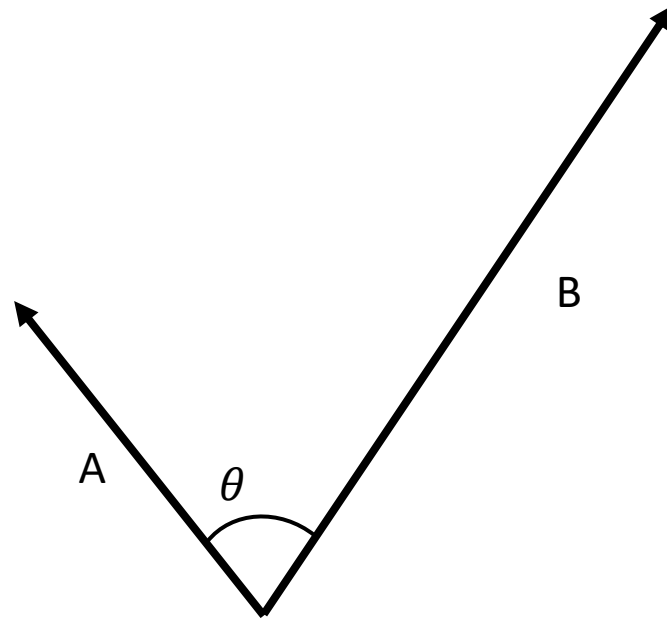
We have two vectors, A and B , each of which has an x and y component

A is (A_x, A_y)

The magnitude of A is $|A|$

B is (B_x, B_y)

The magnitude of B is $|B|$

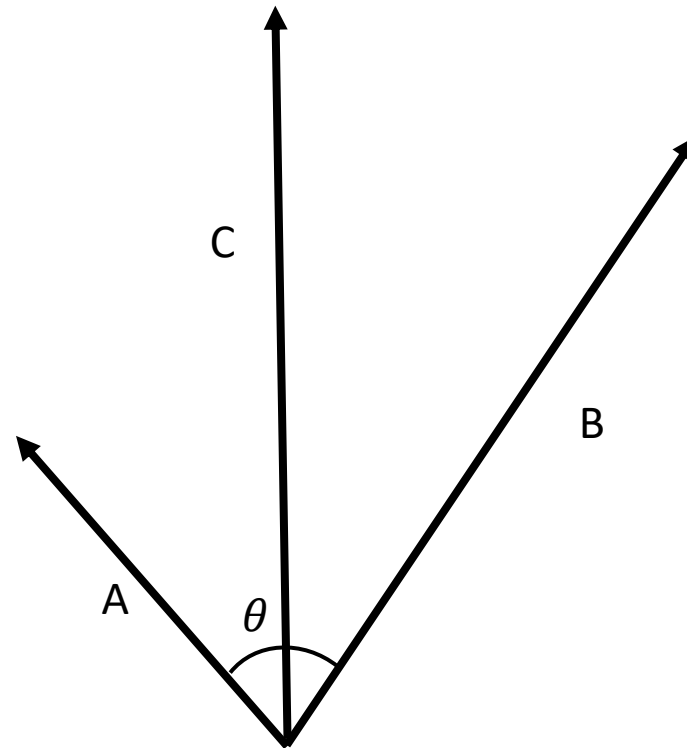


The angle between them is θ

Multiplication – the dot product 1

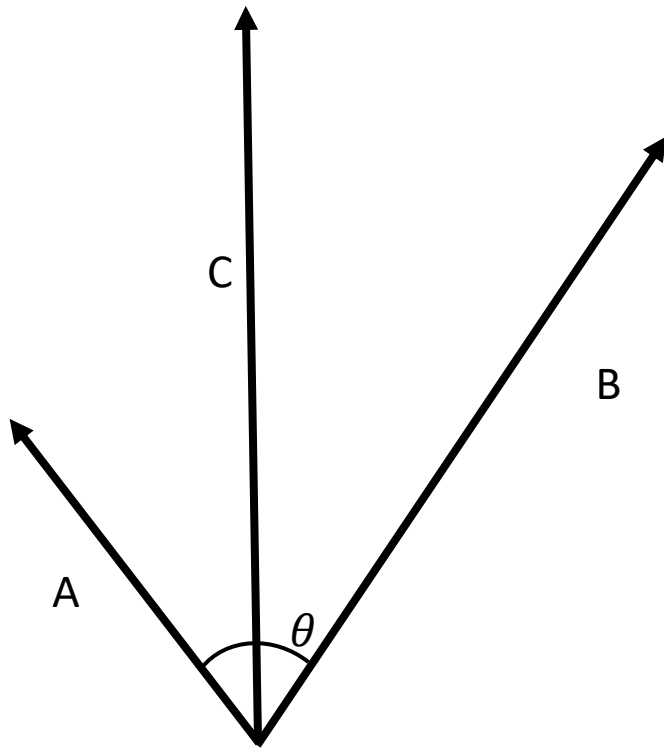
The dot product is a new quantity, C

$\mathbf{C}$ is (C_x, C_y)



The magnitude of C is what interests us

Multiplication – the dot product 2



C is $(A_x \times B_x) + (A_y \times B_y)$

e.g. $A=(-6,7)$, $B=(8, 14)$

$$C = (-6 \times 8) + (7 \times 14) = -48 + 98 = 50$$

C has a magnitude, but no direction – it is a **scalar quantity**

Multiplication – the dot product 3

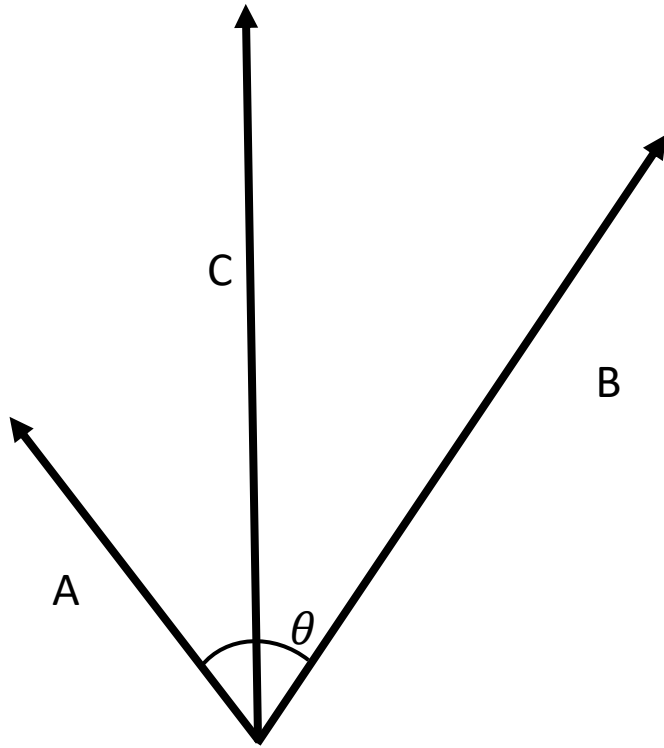
We can do this another way if we know θ

$$a \cdot b = |C| = |A||B|\cos(\theta)$$

e.g. $|A|=9.22, |B|=16.12, \theta=60$

$$C = 9.22 \times 16.12 \times 0.5 = -48 + 98 = 50$$

We can also use this relation to find theta!



Example of dot product 1

Find the magnitude of C

$$|\mathbf{C}| = \mathbf{A} \cdot \mathbf{B}$$

$$\mathbf{A}=(2,3) \quad \mathbf{B}=(-1,5)$$

We know that:

$$|\mathbf{C}| = (2 \times -1) + (3 \times 5) = -2 + 15 = 13$$

Example of dot product 2

We can also find theta:

$$\mathbf{A}=(2,3) \quad \mathbf{B}=(-1,5)$$

$$a \cdot b = |a||b|\cos(\theta)$$

$$13 = |a||b|\cos(\theta)$$

$$13 = \sqrt{2^2 + 3^2} \times \sqrt{-1^2 + 5^2} \times \cos\theta$$
$$\theta = \cos^{-1} \frac{a \cdot b}{|a||b|} = \cos^{-1} \left(\frac{13}{3.6 \times 5.1} \right) = 45^\circ$$

Special case

We know that the dot product of two vectors is given by: $a \cdot b = |A||B|\cos(\theta)$

So what happens when we have 90° between our vectors?

$$\cos(90) = 0$$

So we get a zero result for any perpendicular vectors – this makes it especially useful to have our vectors in Cartesian form

e.g. $\mathbf{a} = 2i + 0j$ $\mathbf{b} = 0i + 1j$

One extends along the x axis, the other on the y axis

$$a \cdot b = (2 \times 0) + (0 \times 1) = 0$$

Uses of dot product

The dot product of **force** and **velocity** is **Power**

The dot product of **displacement** and **force** is **Work**

The dot product of **angle of solar rays** and **angle of solar panels** is **energy absorbed**



[Further help here](#)

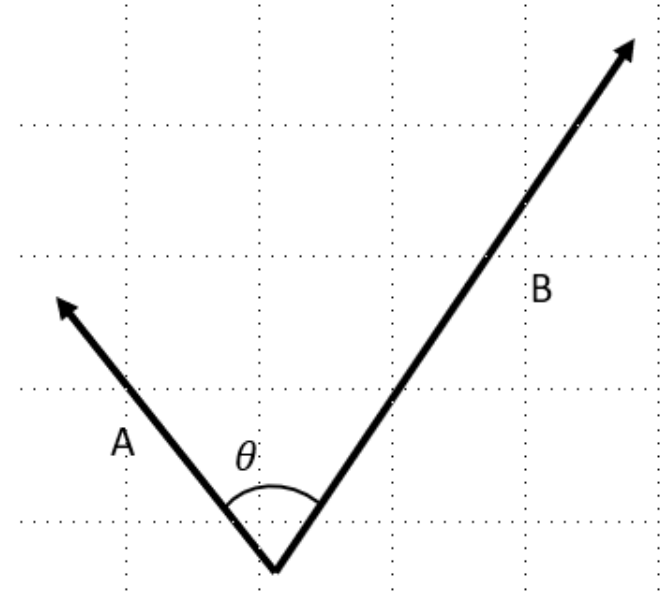
Cross product

We can also multiply vectors in a way that gives another vector

We define $\mathbf{a} \times \mathbf{b}$ as the cross product

This new vector has a magnitude of $ab \sin \theta$ and acts in a direction normal to both $\mathbf{a}$ and $\mathbf{b}$

Here, $\mathbf{a} \times \mathbf{b}$ would go into the screen and $\mathbf{b} \times \mathbf{a}$ would come out of it



Cross product magnitude

There is another way to approach this, but we'll save that until we cover matrices

For now, we use the relation: $|\mathbf{a} \times \mathbf{b}| = ab \sin \theta$

So if $\mathbf{a} = (2,3)$, $\mathbf{b} = (1,4)$ then $|\mathbf{a} \times \mathbf{b}|$:

$$a = (2,3) \times b = (1,4) = \sqrt{2^2 + 3^2} \sqrt{1^2 + 4^2} = \sqrt{13} \sqrt{17} \approx 14.87$$

What about $\sin \theta$?

Finding theta

We did this easily for the dot product:

$$a \cdot b = |a||b|\cos(\theta) \quad \text{so} \quad \theta = \cos^{-1} \left(\frac{a \cdot b}{|a||b|} \right)$$

We have: $\mathbf{a} = (2,3)$, $\mathbf{b} = (1,4)$

$$\theta = \cos^{-1} \left(\frac{2(1) + 3(4)}{14.87} \right) = 19.7$$

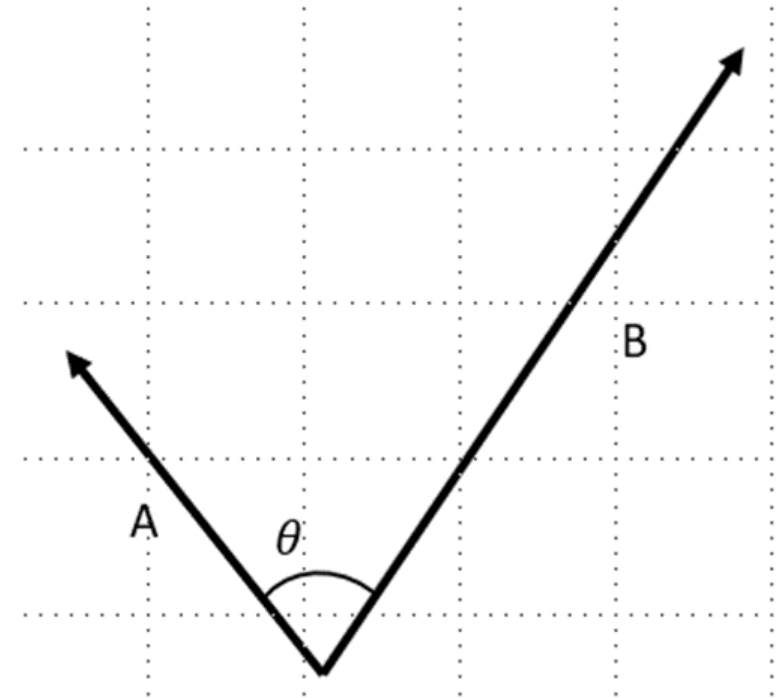
$$|\mathbf{a} \times \mathbf{b}| = ab \sin \theta = 14.87 \sin 19.7 = 11.18$$

Cross product direction

If you are left-handed, tremble with fear

There are many variations of the right hand rule, and many explanations. Like SOHCAHTOA, you need to find your own key to memorising it

If you first, index, finger points to $\mathbf{a}$ and your second finger points to $\mathbf{b}$, your thumb should point to $\mathbf{a} \times \mathbf{b}$



Alternative method for cross products

We can also find the cross product as a determinant:

$$\mathbf{a} = 3\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}, \quad \mathbf{b} = 4\mathbf{i} + 9\mathbf{j} + 2\mathbf{k}$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 4 & 7 \\ 4 & 9 & 2 \end{vmatrix}$$

Determinants

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} i & j & k \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

$$\mathbf{a} \times \mathbf{b} = i(a_y b_z - a_z b_y) - j(a_x b_z - a_z b_x) + k(a_x b_y - a_y b_x)$$

Determinants example

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} i & j & k \\ 3 & 4 & 7 \\ 4 & 9 & 2 \end{vmatrix}$$

$$\mathbf{a} \times \mathbf{b} = i(4 \times 2 - 9 \times 7)$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} i & j & k \\ 3 & 4 & 7 \\ 4 & 9 & 2 \end{vmatrix}$$

$$-j(3 \times 2 - 7 \times 4)$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} i & j & k \\ 3 & 4 & 7 \\ 4 & 9 & 2 \end{vmatrix}$$

$$+k(3 \times 9 - 4 \times 4)$$

Answer

Find the cross product as a determinant:

$$\mathbf{a} = 3i + 4j + 7k, \quad \mathbf{b} = 4i + 9j + 2k$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} i & j & k \\ 3 & 4 & 7 \\ 4 & 9 & 2 \end{vmatrix}$$

$$\mathbf{a} \times \mathbf{b} = -55i + 22j - 11k$$

Uses of cross product

Screwdriver uses forces in the x-y plane (rotation) to cause movement in or out through the z plane

The force experienced by a wire carrying electric current in a magnetic field is the cross product of the current and field vectors

And many more...